



Operating Instructions

Diesel Engine

12V2000Gx5

12V2000Gx5-TB

16V2000Gx5

16V2000Gx5-TB

18V2000Gx5

18V2000Gx5-TB

MS15020/08E

Engine model	kW/cyl.	Application group
12V2000G25	52.9 kW/cyl.	3D, emergency standby power, IFN
12V2000G25-TB	52.9 kW/cyl.	3D, emergency standby power, IFN
12V2000G45	65 kW/cyl.	3D, emergency standby power, IFN
12V2000G45-TB	65 kW/cyl.	3D, emergency standby power, IFN
12V2000G65	63.8 kW/cyl.	3D, emergency standby power, IFN
12V2000G65-TB	63.8 kW/cyl.	3D, emergency standby power, IFN
12V2000G85	74.2 kW/cyl.	3D, emergency standby power, IFN
12V2000G85-TB	74.2 kW/cyl.	3D, emergency standby power, IFN
16V2000G25	55.6 kW/cyl.	3D, emergency standby power, IFN
16V2000G25-TB	55.6 kW/cyl.	3D, emergency standby power, IFN
16V2000G45	63.1 kW/cyl.	3D, emergency standby power, IFN
16V2000G45-TB	63.1 kW/cyl.	3D, emergency standby power, IFN
16V2000G65	60.9 kW/cyl.	3D, emergency standby power, IFN
16V2000G65-TB	60.9 kW/cyl.	3D, emergency standby power, IFN
16V2000G85	69.7 kW/cyl.	3D, emergency standby power, IFN
16V2000G85-TB	69.7 kW/cyl.	3D, emergency standby power, IFN
18V2000G65	61.1 kW/cyl.	3D, emergency standby power, IFN
18V2000G65-TB	61.1 kW/cyl.	3D, emergency standby power, IFN
18V2000G85	72.8 kW/cyl.	3D, emergency standby power, IFN
18V2000G85-TB	72.8 kW/cyl.	3D, emergency standby power, IFN

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

© Copyright Rolls-Royce Solutions

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of Rolls-Royce Solutions. This particularly applies to its reproduction, distribution, editing, translation, microfilming and storage and/or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. Rolls-Royce Solutions reserves the right to change, delete or supplement the information provided as and when required.

California Proposition 65

This warning applies only to the State of California, USA, in compliance with Title 27 California Code of Regulations, Article 6, Clear and Reasonable Warnings.



WARNING: Breathing diesel engine exhaust exposes you to chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

- Always start and operate the engine in a well-ventilated area.
- If in an enclosed area, vent the exhaust to the outside.
- Do not modify or tamper with the exhaust system.
- Do not idle the engine except as necessary.

For more information go to www.P65warnings.ca.gov/diesel.

(→ www.P65warnings.ca.gov/diesel)

Table of Contents

1	Preface		
1.1	Preface	7	
2	Safety		
2.1	Important provisions for all products	8	
2.2	Important requirements for Power Generation application products	9	
2.3	Intended use of all products	11	
2.4	Personnel and organizational requirements	12	
2.5	Safety regulations for commissioning and operation	13	
2.6	Safety regulations for assembly, maintenance, and repair work	15	
2.7	Fire and environmental protection, indirect materials	19	
2.8	Standards for warning notices in the text and highlighted information	22	
3	Transport		
3.1	Transport	23	
4	General Information		
4.1	Engine side and cylinder designations	24	
4.2	Engine overview	25	
4.3	Sensors and actuators - Overview	27	
5	Technical Data		
5.1	Product data 12V/16V/18V2000Gx5, optimized fuel consumption	29	
5.2	Product data 12V/16V/18V2000Gx5-TB, optimized fuel consumption	33	
5.3	Product data 12/16/18V2000Gx5, emissions-optimized	38	
5.4	Product data 12/16/18V2000Gx5-TB, emissions-optimized	42	
5.5	Firing order	46	
5.6	Engine - Main dimensions	47	
6	Operation		
6.1	Preparations for putting into operation after extended out-of-service periods (>3 months)	48	
6.2	Preparations for putting into operation after scheduled out-of-service-period	49	
6.3	Starting the engine in manual mode (test run)	50	
6.4	Re-starting the engine following an automatic safety shutdown	51	
6.5	Safety system - Override	52	
6.6	Operational checks	53	
6.7	Emission label - Check	54	
6.8	Engine - Stopping in manual mode (test run)	55	
6.9	Engine - Emergency stop	57	
6.10	After stopping the engine - Engine remains ready for operation	58	
6.11	After shutting down engine - Putting engine out of operation	59	
7	Maintenance		
7.1	Maintenance task reference table [QL1] for Non-EPA regions	60	
7.2	Maintenance task reference table [QL1] for EPA regions	61	
8	Troubleshooting		
8.1	Fault indication on SAM display - Genset applications	62	
8.2	Troubleshooting	86	
9	Task Description		
9.1	Engine	89	
9.1.1	Engine - Barring manually	89	
9.1.2	Engine - Barring with starting system	90	
9.1.3	Engine - Test run	91	
9.2	Cylinder Liner	92	
9.2.1	Cylinder liner - Endoscopic examination	92	
9.2.2	Cylinder liner - Instructions and comments on endoscopic and visual examination	94	
9.3	Crankcase Breather	96	
9.3.1	Crankcase breather - Oil separator replacement, diaphragm check and replacement	96	
9.4	Valve Drive	98	
9.4.1	Valve clearance - Check and adjustment	98	
9.4.2	Cylinder head cover - Removal and installation	101	
9.5	Injection Pump / HP Pump	102	
9.5.1	Injection pump - Replacement	102	
9.5.2	Injection pump - Removal and installation	103	
9.6	Injection Valve / Injector	106	
9.6.1	Injector - Replacement	106	
9.6.2	Injector - Removal and installation	107	

9.7 Fuel System	113	9.14.5 Coolant pump - Relief bore check	146
9.7.1 HP fuel line - Pressure pipe neck replacement	113	9.15 Low-Temperature Circuit	147
9.7.2 Fuel - Draining	116	9.15.1 Charge-air coolant - Filling	147
9.7.3 Fuel pressure relief valve - Removal and installation	118	9.15.2 Charge-air coolant - Draining	149
9.7.4 Fuel system - Venting	119	9.15.3 Charge-air coolant - Change	151
9.8 Fuel Filter	122	9.15.4 Charge-air coolant level - Check	152
9.8.1 Fuel filter - Replacement	122	9.16 Battery-Charging Generator	154
9.8.2 Fuel prefilter - Differential pressure gauge check and adjustment	123	9.16.1 Battery-charging generator drive - Drive-belt check and adjustment	154
9.8.3 Fuel prefilter - Draining	124	9.16.2 Battery-charging generator drive - Drive belt replacement	156
9.8.4 Fuel prefilter - Flushing	126	9.17 Fan Drive	157
9.8.5 Fuel prefilter - Cleaning	129	9.17.1 Fan drive - Drive-belt check and adjustment	157
9.8.6 Fuel prefilter - Filter element replacement	130	9.17.2 Fan drive - Drive belt replacement	160
9.9 Charge-Air Cooling General, Left-Hand Side	132	9.18 Wiring (General) for Engine/Gearbox/Unit	161
9.9.1 Intercooler - Checking condensate drain for coolant discharge and obstructions	132	9.18.1 Engine cabling - Check	161
9.10 Air Filter	133	9.19 Accessories for (Electronic) Engine Governor / Control System	162
9.10.1 Air filter - Replacement	133	9.19.1 Engine governor and connectors - Cleaning	162
9.10.2 Air filter - Removal and installation	134	9.19.2 Engine governor plug connections - Check	163
9.11 Air Intake	135	9.19.3 Engine governor ECU 7 - Removal and installation	164
9.11.1 Service indicator - Signal ring position check	135	9.19.4 Engine control unit - Reading out fault memory	165
9.12 Lube Oil System, Lube Oil Circuit	136	10 Appendix A	
9.12.1 Engine oil level - Check	136	10.1 Abbreviations	166
9.12.2 Engine oil - Change	137	10.2 Contact person/Service partner	170
9.13 Oil Filtration / Cooling	139	11 Appendix B	
9.13.1 Engine oil filter - Replacement	139	11.1 Standard Tools and Special Tools	171
9.14 Coolant Circuit, General, High-Temperature Circuit	140	11.2 Index	178
9.14.1 Engine coolant - Level check	140		
9.14.2 Engine coolant - Change	142		
9.14.3 Engine coolant - Draining	143		
9.14.4 Engine coolant - Filling	144		

1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at:<http://www.mtu-solutions.com>

2.2 Important requirements for Power Generation application products

Project-specific operational environment

During integration and use of the diesel engine, all currently valid and relevant standards and guidelines on safety and specified operation must be implemented and observed. This applies to persons responsible for:

- The installation of the engine
- The design of the engine-generator set
- The installation of the engine-generator set

The relevant standards are, in particular:

- DIN EN 1679-1
- EN ISO 8528-13

Residual hazards

In the event of a defect, the diesel engine can unexpectedly have insufficient or no drive power at all. We therefore recommend a redundant design of the engine-generator set system in safety-critical applications.

Protection against hot surfaces

Combustion engines and their auxiliary components have hot surfaces (e.g. exhaust system, exhaust turbochargers, exhaust aftertreatment system, engine oil system, cooling system), which can cause injury to persons who approach or touch them (danger of burns).

- In operation
- During the cooling-down phase after operation

The plant integrator and operator must assess, based on spatial or other plant conditions, whether the danger potential usually associated with the state of technology is exceeded, e.g.:

- Particularly narrow maintenance spaces
- Particularly exposed position of hot surfaces

In such cases, the plant integrator or operator must equip the plant with the necessary and suitable safety devices to reliably exclude the possibility of personal injury, e.g.:

- Heat shields
- Insulation

The operator must ensure that the plant is not operated without these protective devices. Operating and maintenance personnel must be informed about the danger of hot surfaces and the prohibition of operation without the protective devices.

Noise

Noise protection measures that comply with statutory requirements must be implemented by the plant integrator and operator.

Chemical resistance

Only use hoses, pipes, fittings, etc. that can tolerate the fluids and lubricants in accordance with the Fluids and Lubricants Specifications. Only use options that are resistant to the fluids and lubricants in accordance with the Fluids and Lubricants Specifications.

Electrical devices

The IP protection class of the electrical devices required by the manufacturer must be observed.

Hoses

For vibration decoupling, where possible only use hoses or rubber sleeves. Install devices in low-vibration areas and not on the engine. Hoses must be installed such that they do not chafe and are not under tension.

Incorrect synchronization

Load application with incorrect synchronization (generator connection with phase offset) can cause considerably high torque pulses. The plant integrator must design the control system such that the possibility of a load application with incorrect synchronization is excluded.

Emergency stop

The diesel engine has inputs for Stop and Emergency stop. The emergency-stop function must be executable in the whole plant independently of the operation stop. This is the only way to ensure that a safe shutdown of fuel injection takes place on a diverse redundant path and thus results in engine standstill. Through an appropriate design of the engine-generator set or whole plant, the operator of the engine must guarantee that safe, non-reversible stopping of the engine is as fast as possible, including all of its dangerous system components, when the emergency-stop actuating element is activated. In particular, it must always be guaranteed that machine shutdown can not be prevented or delayed once the command for shutdown has been issued. The engine shutdown command must always have priority over the start command.

If machines or machine components are designed to interact, they must be designed and built such that all devices can shut down. This relates to:

- Emergency-stop command devices for the machine itself (engine, engine-generator set)
- All connected devices insofar as their continued operation could result in a hazard

Important

Engines with exhaust aftertreatment: When the emergency stop is actuated on engine with an exhaust aftertreatment system, note that the emergency-stop function also interrupts the reducing agent supply. This can result in overheating of the reducing agent dosing units. For details and further information, see Operating Instructions.